

Use of these files to explore MOPAC Version 1.14

Preamble

MOPAC 1.14 is the earliest copy of MOPAC known to exist. It was written entirely at the University of Texas at Austin, in the laboratory of Dr. Michael J. S. Dewar. The original copy, in MOPAC_1_Source_Dated_1983.zip no longer compiles using modern compilers. A version of this program that has been modified so that it will run using modern compilers, named MOPAC-1, will allow the original code to be explored easily.

Only a very limited range of functionality is supported. This is a deliberate attempt to focus attention on the code itself, and to discourage its use in a release mode.

Some changes were made to the original code from 1983 in order to allow it to be compiled using modern compilers. Only essential changes were made, these fall into the following groups:

Machine-specific lines commented out, example: In “second.for” the line “**CALL DEFINE_EVENT_FLAG_CLUSTER**”, has been commented out. This call was used only by a specific type of hardware, and did not affect the normal running of the program. In each case, the modern equivalent has NOT been implemented in compliance with the intent to minimize the changes.

Lines added: An example of a line that had to be added can be seen in “wrtkey.for,” where the line: “**double precision, external :: reada**” has been added. This line allows the modern compiler to recognize that “reada” is a function that will be defined elsewhere. All lines added are in lower-case, to allow them to be easily identified; the original MOPAC was written entirely in upper-case.

Lines modified: Some constructions that had been allowed in 1983 are no longer allowed. An example in “linmin.for” is as follows:

```
CALL COMPMFG(XPARAM, .TRUE., PHI(2),FULSCF,GRAD,SIGMA)
```

In this call, the last argument “SIGMA” is not defined, but in the subroutine this argument is logical. So to comply with modern conventions, the last argument was changed to .false., as shown here:

```
CALL COMPMFG(XPARAM, .TRUE., PHI(2),FULSCF,GRAD,.false.)
```

Comment lines: Many comment line characters were deleted when the original ZIP file was made. These would have been either “C” or “*” These have all been replaced with upper-case “C”.

Compiler modifications:

The Windows version of the Intel Fortran compiler was used within Visual Studio. When run in debug mode, this provides the best view of how MOPAC-1 works. Two modifications were made to the default Visual Studio:

In the Fortran Command Line add /Qsave. The early compiler automatically SAVE'd everything; now variables that need to be saved are normally defined locally to be save'd.

In the Fortran Command Line add /Qinit:zero. The early compiler automatically initialized everything; this option does the same.